EXPERIMENT – 11

Hiding and Extracting a Text File behind an Image / Audio File using Command Prompt

Aim

To study the steps for hiding and extracting any text file behind an image file or an audio file using the Command Prompt.

Procedure

- Hiding the text file in an image: Any file such as .rar, .jpg or .txt can be merged inside another file. Using the Command Prompt, a text file can be hidden inside an image file.
- Copy the file to be hidden (for example sample.txt) and the carrier image (for example sample1.jpg) into the same folder.
- Open the Command Prompt (Ctrl+R → cmd) and navigate to that folder using the cd command.
- Merge the two files with the binary copy command: `copy /b sample1.jpg + sample.txt result.jpg`
- The image file and the text file are combined into the output file result.jpg, which still opens as a normal image file.
- Extracting the hidden file: To retrieve the hidden data, locate result.jpg, right-click it and open it with Notepad – the hidden content is found at the end of the file.
- Hiding a RAR archive: In the same way a RAR archive can be hidden behind an image: `copy /b sample1.jpg + sample.rar result.jpg`, and the archive is opened by opening result.jpg with WinRAR.
- Hiding a ZIP in an audio file: A ZIP archive can be hidden behind an audio file: `copy /b sound.mp3 + new.zip newfile.mp3`. The data is recovered either by changing the extension from .mp3 to .zip or by opening newfile.mp3 with a ZIP program.

- On Linux the same merge can be performed with: `cat sound.mp3 secret.zip > newfile.mp3`

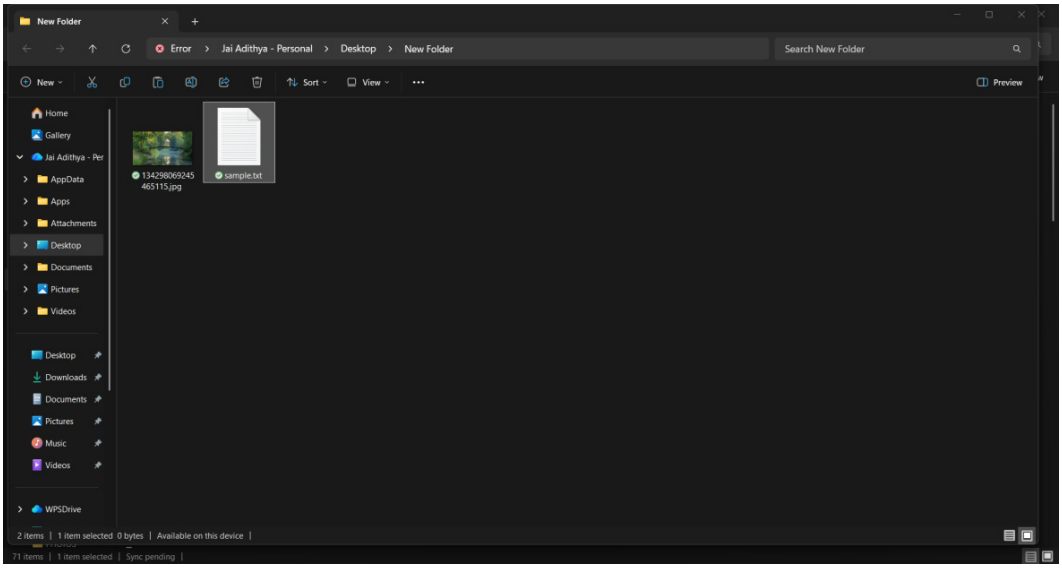


Fig. 11.1 Placing the text file and the carrier image in a folder
Timestamp: 14-10-2026, 20:30:41

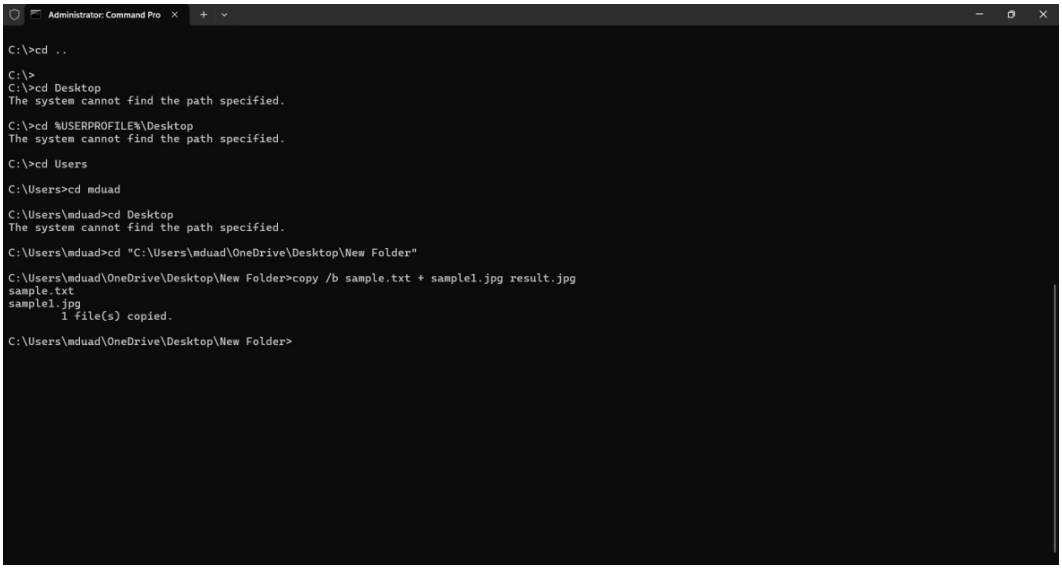


Fig. 11.2 Command Prompt – merging the text file with the image (copy /b)
Timestamp: 14-10-2026, 20:37:19

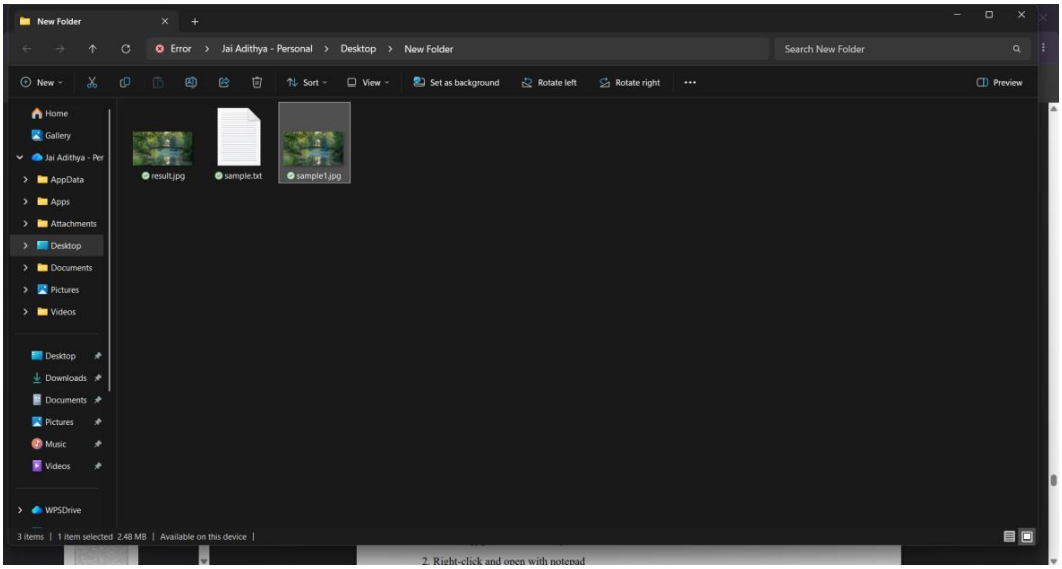


Fig. 11.3 The output image result.jpg generated after the merge
Timestamp: 14-10-2026, 20:37:39

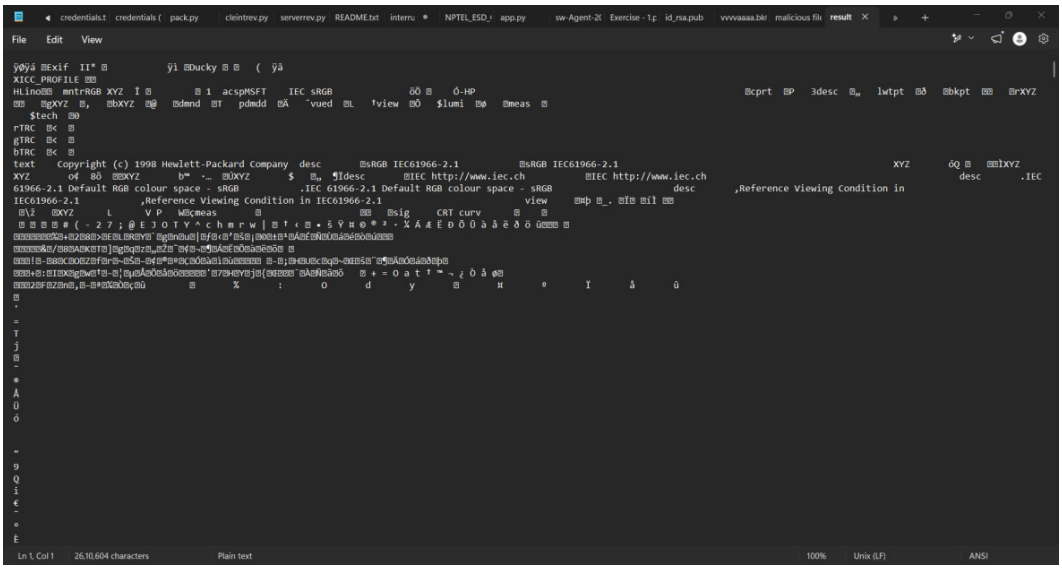


Fig. 11.4 Opening result.jpg with Notepad to retrieve the hidden text
Timestamp: 14-10-2026, 20:39:23

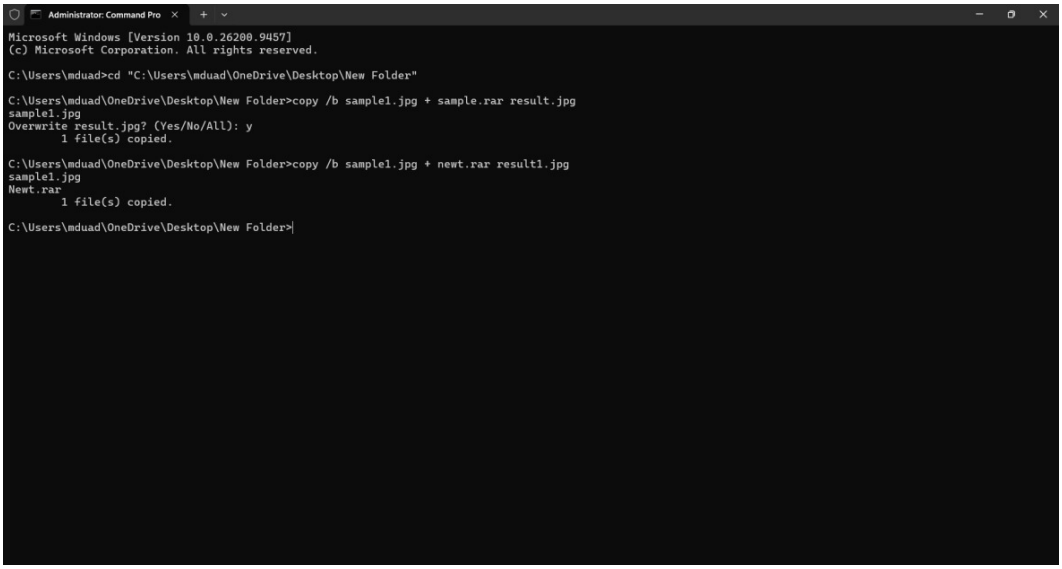


Fig. 11.5 Command Prompt – hiding a RAR archive behind the image
Timestamp: 14-10-2026, 20:44:24

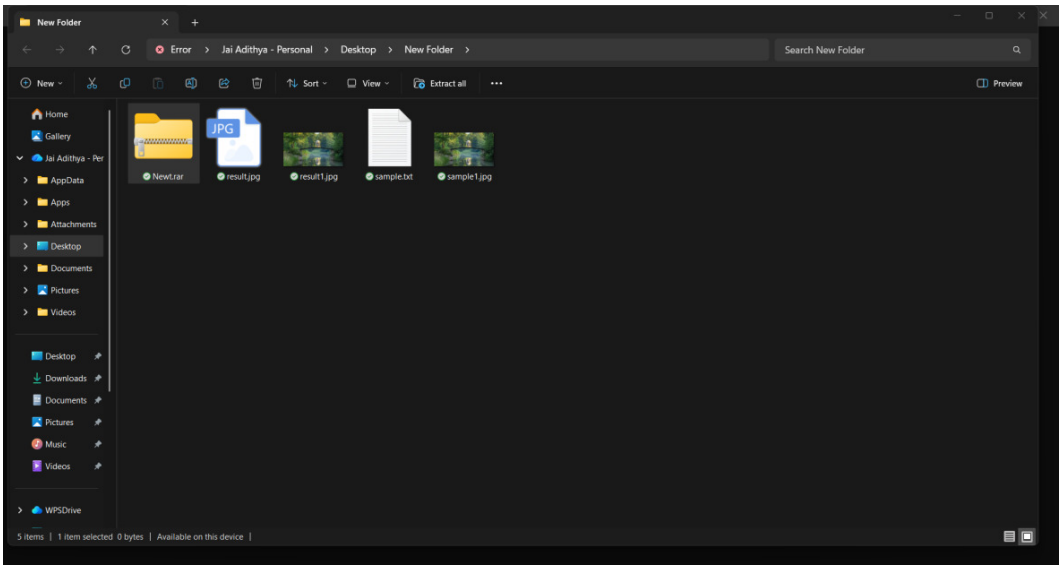


Fig. 11.6 The output image containing the hidden RAR archive
Timestamp: 14-10-2026, 20:45:23

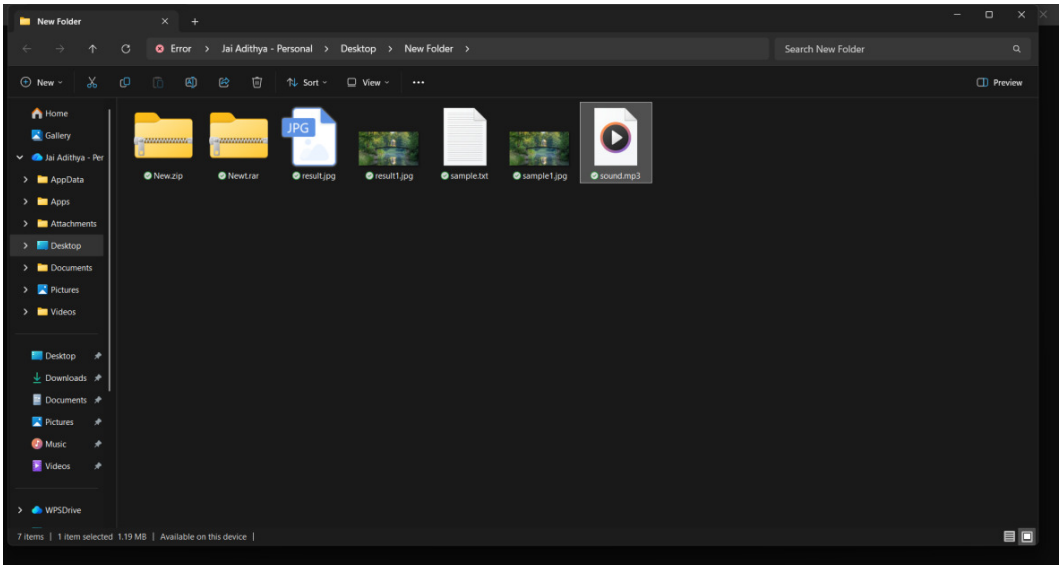


Fig. 11.7 Files prepared for hiding data behind an audio file
Timestamp: 14-10-2026, 20:48:55

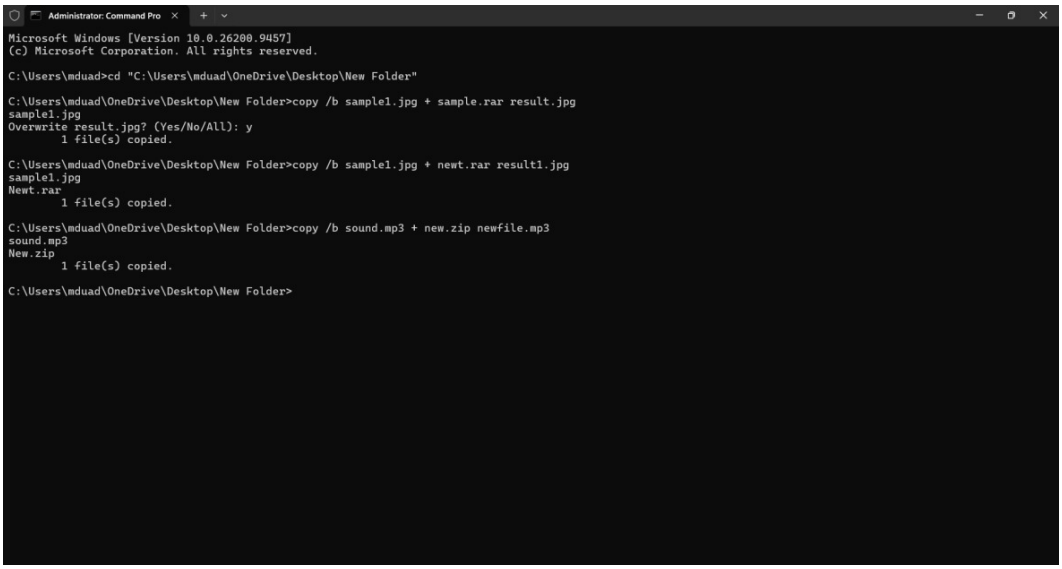


Fig. 11.8 Command Prompt – merging the ZIP archive with the audio file
Timestamp: 14-10-2026, 20:49:31

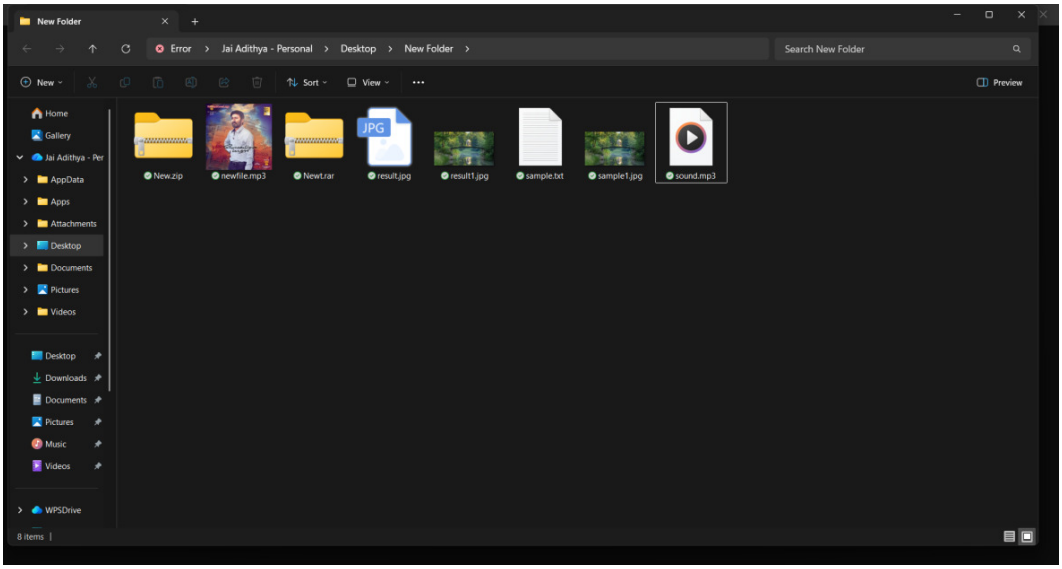


Fig. 11.9 The output audio file newfile.mp3 containing the hidden ZIP archive
Timestamp: 14-10-2026, 20:49:43

Result

Thus, a text file, a RAR archive and a ZIP archive were hidden behind an image file and an audio file using the Command Prompt, and the hidden content was retrieved successfully. The output files continued to open normally as image and audio files.